A deep learning approach to the probabilistic numerical solution of path-dependent partial differential equations

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Abstract

Recent work on Path-Dependent Partial Differential Equations (PPDEs) has shown that PPDE solutions can be approximated by a probabilistic representation, implemented in the literature by the estimation of conditional expectations using regression. However, a limitation of this approach is to require the selection of a basis in a function space. In this paper, we overcome this limitation by the use of deep learning methods, and we show that this setting allows for the derivation of error bounds on the approximation of conditional expectations. Numerical examples based on a two-person zero-sum game, as well as on Asian and barrier option pricing, are presented. In comparison with other deep learning approaches, our algorithm appears to be more accurate, especially in large dimensions.

Keywords: Path-dependent partial differential equations (PPDEs), deep neural networks, numerical methods for PPDEs.

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1 Introduction

Fully nonlinear PPDEs of the form

$$\begin{cases} \partial_t u(t,\omega) + b(t,\omega)\cdot\partial_\omega u(t,\omega) + \dfrac{1}{2}\sigma\sigma^\top(t,\omega) : \partial_{\omega\omega} u(t,\omega) + F\left(\cdot, u, \partial_\omega u, \partial^2_{\omega\omega} u\right)(t,\omega) = 0, \\ u(T,\omega) = g(\omega), \end{cases} \tag{1.1}$$

have been introduced in [Pen11] and their well-posedness in the sense of viscosity solutions have been studied in [EKTZ14, ETZ16a, ETZ16b]. Here, ω is in the set of $\mathbb{R}^d$-valued continuous paths, $b(t,\omega)$ is $\mathbb{R}^d$-valued, $\sigma(t,\omega)$ takes values in the space of invertible $d\times d$ matrices, and F is a real-valued function on $[0,T]\times\Omega\times\mathbb{R}\times\mathbb{R}^d\times\mathbb{R}^{d\times d}$ satisfying Assumption 1 below. The precise meaning of the partial derivatives $\partial_t u(t,\omega)$, $\partial_\omega u(t,\omega)$ and $\partial_{\omega\omega} u(t,\omega)$, which are connected to the horizontal and vertical derivatives used in the functional Itô formula of [Dup09], will be discussed in Section 2.

PPDEs of the type (1.1) have recently been the object of increased attention due to their ability to model control and pricing problems in a non-Markovian setting, see e.g. [TZ15], [JO19], [VZ19].

Nevertheless, as a large class of PPDE is not analytically solvable, one has to rely on numerical solutions. In [RT17], a probabilistic scheme based on [FTW11] has been proposed, and was proved to converge to the PPDE viscosity solution. However, its practical implementation is far from trivial due to the presence of the conditional expectation. The suggestion of [RT17] to use regression as in [GLW05] relies on a careful basis function choice, which may not always be possible, see the discussion at the end of Section 2.

On the other hand, neural networks methods for fully nonlinear PDEs have been introduced independently in [HJE18] and [SS18] using backward stochastic differential equations and the Galerkin method, respectively. See also [BEJ19], [HPW20], [BBC+21], [PWG21], and [LLP22] for other variants of deep learning-based numerical algorithms, and [EHJK19], [BBH+20], for the recently developed multilevel Picard method.

A deep neural network algorithm for the numerical solution of PPDEs has been proposed in [SZ21] by applying Long Short-Term Memory (LSTM) networks in the framework of the deep Galerkin method for PDEs, see [SS18]. On the other hand, [SVŠS20] propose to combine the LSTM network and the path signature to solve linear PPDEs. A deep

signature algorithm has been recently proposed in [FLZ23] to solve path and state-dependent forward-backward stochastic differential equations (FBSDEs). Unlike regression methods, deep learning algorithms do not rely on the choice of a basis, see also [LL21] for the use of neural networks instead of a regression basis in the pricing of Bermudan options using the Longstaff-Schwarz algorithm.

In this paper, we propose a deep learning approach to the implementation of the probabilistic scheme of [RT17] for the numerical solution of fully nonlinear PPDEs of the form (1.1). The main idea of Algorithm 4.1 is based on the L^2 minimality property of conditional expectations, which allows us to transform the conditional expectation in the probabilistic scheme into an optimization problem (3.4) that can be solved using neural networks.

Additionally, we propose an error bound of the Algorithm 4.1 in Proposition 4.2, which is used to prove its convergence in Theorem 4.4. In Section 5, we detail the implementation of our algorithm, followed by numerical examples consisting of a two-person zero-sum game, and Asian and barrier options pricing. For the zero-sum game, Algorithm 4.1 appears more accurate than the deep learning algorithm of [SZ21] in dimensions 1 and 10, and more precise than [RT17] in dimension $d = 100$, see Table 1. For Asian options, our results also improve on [SVŠS20] and [SZ21] in dimension $d = 10$, see Table 2. In dimension $d = 100$ for barrier options, our algorithm is also more precise than [RT17], and more precise than [SVŠS20] and [SZ21] in dimension $d = 10$, see Table 3. In comparison with [SVŠS20], our method avoids the computation of Hessian matrices. However, our numerical simulations are restricted in time to $T = 0.1$, for which our algorithm performs optimally.

The Python codes designed for our numerical experiments are available at `https://github.com/nguwijy/deep_ppde`.

This paper is organized as follows. The necessary preliminaries on PPDEs are stated in Section 2, and the probabilistic schemes introduced in [FTW11] for PDEs and in [RT17] for PPDEs are reviewed in Section 3. In Section 4 we present our main deep learning algorithm, and derive its error bound and convergence respectively in Proposition 4.2 and Theorem 4.4. Numerical implementation and examples are presented in Section 5.

2 Viscosity solutions of path-dependent PDEs

Fix $T > 0$, $x_0 \in \mathbb{R}^d$, let $\Omega = \mathcal{C}_{x_0}([0,T];\mathbb{R}^d)$ denote the set of $\mathbb{R}^d$-valued continuous paths ω starting at $\omega_0 = x_0$ and let $\Theta := [0,T] \times \Omega$. We let $B = (B_t)_{t\in[0,T]}$ denote the $\mathbb{R}^d$-valued canonical process on Ω, while $\mathbb{F} = (\mathcal{F}_t)_{0\le t\le T}$ denotes the canonical filtration generated by $(B_t)_{t\in[0,T]}$, and $\mathbb{P}_0$ is the Wiener measure on $(\Omega, \mathbb{F})$. A natural metric $\mathbf{d}$ on Θ is defined as

$$\mathbf{d}((t,\omega),(t',\omega')) = |t-t'| + \|\omega_{t\wedge\cdot} - \omega'_{t'\wedge\cdot}\|, \qquad (t,\omega),(t',\omega') \in \Theta,$$

where $\|\omega\| := \sup_{t\in[0,T]} \|\omega_t\|_d$, $\omega \in \Omega$, and $\|\cdot\|_d$ denotes the Euclidean norm on $\mathbb{R}^d$. Next, we define $\mathcal{P}$ as the set of probability measures $\mathbb{P}$ such that the canonical process

$$B_t = A^{\mathbb{P}}_t + M^{\mathbb{P}}_t = \int_0^t \mu^{\mathbb{P}}_s ds + M^{\mathbb{P}}_t, \qquad t \in [0,T],$$

is a semimartingale, where $(A^{\mathbb{P}})_{t\in[0,T]}$ is a finite variation process and $(M^{\mathbb{P}})_{t\in[0,T]}$ is a continuous martingale, so that the quadratic variation

$$\langle M^{\mathbb{P}}\rangle_t = \int_0^t a^{\mathbb{P}}_s ds, \qquad t \in [0,T],$$

taking values in the space $\mathbb{S}^d$ of $d \times d$ symmetric matrices is absolutely continuous with respect to the Lebesgue measure on $[0,T]$, and

$$\sup_{t\in[0,T]} \|\mu^{\mathbb{P}}_t\|_d \le L, \qquad \frac{1}{2}\sup_{t\in[0,T]} \mathrm{Tr}\big(a^{\mathbb{P}}_t\big) \le L, \quad \mathbb{P}\text{-a.s.},$$

where $L > 0$ is fixed throughout the paper. We also denote by $v_1 \cdot v_2$ the dot product of $v_1, v_2 \in \mathbb{R}^d$, let $\boldsymbol{I}_d$ be the identity matrix in $\mathbb{S}^d$, let $A : B = \mathrm{Tr}(AB)$, and let $\mathbf{1}_d = (1,\dots,1)$ represent the all-ones vector in $\mathbb{R}^d$. We refer to Theorem 2.4 in [ETZ16a] for the following definition which makes sense of the partial derivatives on Θ $\partial_\omega u(t,\omega)$, $\partial_{\omega\omega} u(t,\omega)$, in the PPDE (1.1), which are connected to the path derivatives used in functional Itô formulas, see [Dup09], or Proposition 6 in [Pri97].

Definition 2.1 *We say that $u \in C^{1,2}(\Theta)$ if $(t,\omega) \mapsto u(t,\omega) \in \mathbb{R}$ is continuous on (Θ, d) and there exists continuous processes $(t,\omega) \mapsto \partial_t u \in \mathbb{R}$, $(t,\omega) \mapsto \partial_\omega u(t,\omega) \in \mathbb{R}^d$, $(t,\omega) \mapsto \partial_{\omega\omega} u(t,\omega) \in \mathbb{S}^d$, continuous on (Θ, d), and such that*

$$du(t,\omega) = \partial_t u(t,\omega)dt + \frac{1}{2}\partial_{\omega\omega} u(t,\omega) : d\langle B\rangle_t + \partial_\omega u(t,\omega)\cdot dB_t, \quad \mathbb{P}\text{-a.s., for all } \mathbb{P} \in \mathcal{P}.$$

As in e.g. [RT17] and [FTW11], we assume that the coefficients of (1.1) satisfy the following conditions throughout the paper.

Assumption 1 *i)* $\sigma(t,\omega)$ *is invertible in* $\mathbb{S}^d$ *for all* $(t,\omega)\in\Theta$, *and* $b(t,\omega)$, $\sigma(t,\omega)$ *satisfy*

$$\sup_{(t,\omega)\neq(t',\omega')}\frac{\|b(t,\omega)-b(t',\omega')\|_d}{|t-t'|^{1/2}+\|\omega_{t\wedge\cdot}-\omega'_{t\wedge\cdot}\|}+\sup_{(t,\omega)\neq(t',\omega')}\frac{\|\sigma(t,\omega)-\sigma(t',\omega')\|_{d\times d}}{|t-t'|^{1/2}+\|\omega_{t\wedge\cdot}-\omega'_{t\wedge\cdot}\|}<\infty,$$

where $\|\cdot\|_{d\times d}$ *denotes the Frobenius norm on* $\mathbb{S}^d$.

ii) $\sigma^{-1}(t,\omega)$ *is bounded in* $\mathbb{S}^d$, *uniformly in* $(t,\omega)\in\Theta$.

iii) $\omega\mapsto g(\omega)$ *is bounded and Lipschitz on* Ω,

iv) $F(t,\omega,u,z,\gamma)$ *is continuous in* $(t,\omega,u,z,\gamma)\in\Theta\times\mathbb{R}\times\mathbb{R}^d\times\mathbb{S}^d$.

v) $F(t,\omega,u,z,\gamma)$ *is Lipschitz with respect to* $(\omega,u,z,\gamma)\in\Omega\times\mathbb{R}\times\mathbb{R}^d\times\mathbb{S}^d$ *uniformly in* $t\in[0,T]$, *and* $\sup_{(t,\omega)\in\Theta}|F(t,\omega,0,0,0)|<\infty$.

vi) $F(t,\omega,u,z,\gamma)$ *is elliptic, i.e.* $\gamma\mapsto F(t,\omega,u,z,\gamma)$ *is non-decreasing in* $\gamma\in\mathbb{S}^d$ *for the positive semidefinite order* $\leq_{\rm psd}$.

vii) $F(t,\omega,u,z,\gamma)$ *satisfies* $\dfrac{\partial F}{\partial\gamma}(t,\omega,u,z,\gamma)\leq_{\rm psd}(\sigma\sigma^\top)(t\omega)$ *for a.e.* $(t,\omega,u,z,\gamma)\in\Theta\times\mathbb{R}\times\mathbb{R}^d\times\mathbb{S}^d$.

viii) $\dfrac{\partial F}{\partial z}(t,\omega,u,z,\gamma)\in\text{Image}\left(\dfrac{\partial F}{\partial\gamma}(t,\omega,u,z,\gamma)\right)$ *for all* $(t,\omega,u,z,\gamma)\in\Theta\times\mathbb{R}\times\mathbb{R}^d\times\mathbb{S}^d$, *and*

$$\text{ess sup}_{(t,\omega,u,z,\gamma)}\left|\left(\frac{\partial F}{\partial z}\right)^\top\left(\frac{\partial F}{\partial\gamma}\right)^{-1}\frac{\partial F}{\partial z}(t,\omega,u,z,\gamma)\right|<\infty.$$

In general, F may not be smooth enough to ensure the existence of the classical solution for (1.1), hence we rely on the weaker notion of viscosity solution. For this, we will use the shift operations

$$(\omega\otimes_t\omega')_s:=\omega_s\mathbb{1}_{[0,t]}(s)+(\omega'_{s-t}-x_0+\omega_t)\mathbb{1}_{(t,T]}(s),\quad\omega,\omega'\in\Omega,$$

and let, for $u:\Theta\to\mathbb{R}$,

$$u^{t,\omega}(s,\omega'):=u(t+s,\omega\otimes_t\omega'),\quad\omega,\omega'\in\Omega.$$

Next, for u in the set $\mathrm{BUC}(\Theta)$ of all bounded and uniformly continuous functions $u : \Theta \to \mathbb{R}$ on (Θ, d), letting

$$\overline{\mathcal{E}}[\,\cdot\,] := \sup_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}[\,\cdot\,] \quad \text{and} \quad \underline{\mathcal{E}}[\,\cdot\,] := \inf_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}[\,\cdot\,],$$

we define the sets of test functions

$$\overline{\mathcal{A}}u(t,\omega) := \left\{\varphi \in C^{1,2}(\Theta) : (\varphi - u^{t,\omega})_0 = 0 = \sup_{\tau\in\mathcal{T}_{H_\delta}} \overline{\mathcal{E}}\left[(\varphi - u^{t,\omega})_\tau\right]\right\}$$

and

$$\underline{\mathcal{A}}u(t,\omega) := \left\{\varphi \in C^{1,2}(\Theta) : (\varphi - u^{t,\omega})_0 = 0 = \inf_{\tau\in\mathcal{T}_{H_\delta}} \overline{\mathcal{E}}\left[(\varphi - u^{t,\omega})_\tau\right]\right\},$$

$(t,\omega) \in \Theta$, where $H_\delta(\omega') := \delta \wedge \inf\{s \geq 0 : |\omega'_s| \geq \delta\}$, and $\mathcal{T}_{H_\delta}$ is the set of all $\mathbb{F}$-stopping times taking values in $[0, H_\delta]$. The following definition makes sense of the viscosity solution of the PPDE (1.1).

Definition 2.2 *A function $u \in \mathrm{BUC}(\Theta)$ is a viscosity subsolution (resp. supersolution) of the PPDE (1.1) if for any $(t,\omega) \in \Theta$ and for all $\varphi \in \overline{\mathcal{A}}u(t,\omega)$, (resp. $\varphi \in \underline{\mathcal{A}}u(t,\omega)$), we have*

$$\partial_t\varphi(t,\omega) + b(t,\omega)\cdot\partial_\omega\varphi(t,\omega) + \frac{1}{2}\sigma\sigma^\top(t,\omega) : \partial_{\omega\omega}\varphi(t,\omega) + F\left(\,\cdot\,,\varphi,\partial_\omega\varphi,\partial^2_{\omega\omega}\varphi\right)(t,\omega) \geq 0,$$

resp. ≤ 0.

In addition, we say that $u \in \mathrm{BUC}(\Theta)$ is a viscosity solution of the PPDE (1.1) if it is both a viscosity subsolution and a viscosity supersolution of (1.1).

3 Probabilistic numerical solution

In this section, we consider the probabilistic scheme introduced by [FTW11] for PDEs and later generalized to PPDEs in [RT17]. Given $N \geq 1$ and $h := T/N$, consider the random variable

$$X_h^{(t,\omega)} := \begin{pmatrix} x_0 \\ x_0 + b(t,\omega)h + \sigma(t,\omega)B_h \end{pmatrix}, \tag{3.1}$$

where $B_h \sim N(0, h\boldsymbol{I}_d)$ is a d-dimensional Gaussian vector. For any vectorized matrix $y = (x_0, x_1, \ldots, x_i)^{\mathrm{V}} \in \mathbb{R}^{(i+1)d}$ with $0 \leq i \leq N$, we consider the linear interpolation $\overline{y} \in \Omega$ of y defined as

$$\overline{y}_s = \begin{cases} \dfrac{s-kh}{h}x_{k+1} + \left(1 - \dfrac{s-kh}{h}\right)x_k, & s \in [kh, (k+1)h),\ k = 0, 1, \ldots, i-1, \\ x_i, & s \in [ih, T]. \end{cases} \tag{3.2}$$

For $\phi : \Theta \to \mathbb{R}$ a given function, we let $\mathcal{D}_h\phi(t,\omega) = (\phi(t,\omega), \partial_\omega\phi(t,\omega), \partial_{\omega\omega}\phi(t,\omega))$ denote the path partial derivatives of $\phi(t,\omega)$ with respect to ω as in Definition 2.1, rewritten by integration by parts as

$$\mathcal{D}_h\phi(t,\omega) := \mathbb{E}\Big[\phi\big(t+h, \omega \otimes_t \overline{X}_h^{(t,\omega)}\big) H_h(t,\omega) \,\Big|\, \mathcal{F}_t\Big],$$

see [FTW11], [RT17], where $H_h = \big(H_0^h, H_1^h, H_2^h\big)$ are the weights defined by

$$H_0^h := 1, \quad H_1^h := (\sigma^\top)^{-1}\frac{B_h}{h}, \quad H_2^h := (\sigma^\top)^{-1}\frac{B_h B_h^\top - h\boldsymbol{I}_d}{h^2}\sigma^{-1}.$$

As in Equations (2.5) of [FTW11] and (4.11) of [RT17], we let the operator $\mathbb{T}^{t,\omega}$ be defined as

$$\mathbb{T}^{t,\omega}\left[u^h(t+h,\,\cdot\,)\right] := \mathbb{E}\left[u^h\big(t+h, \omega\otimes_t \overline{X}_h^{(t,\omega)}\big) \,\Big|\, \mathcal{F}_t\right] + hF\left(\,\cdot\,, \mathcal{D}_h u^h_{t+h}\right)(t,\omega), \tag{3.3}$$

which is used to construct the probabilistic Euler discretization u^h of u defined inductively as in (2.4) of [FTW11] and § 3 of [RT17], i.e. as the linear interpolation $u^h(t,\omega)$ of the sequence

$$\begin{cases} u^h(Nh,\omega) = g(\omega), \\ u^h(ih,\omega) = \mathbb{T}^{t,\omega}\left[u^h((i+1)h,\,\cdot\,)\right], \qquad i = 0,1,\dots,N-1. \end{cases} \tag{3.4}$$

The convergence of u^h to u as h tends to zero is ensured by the following result, see Theorem 3.9 and Proposition 4.9 in [RT17].

Theorem 3.1 *Under Assumption 1, assume further that the PPDE (1.1) satisfies the comparison principle for viscosity subsolutions and supersolutions, i.e. if v and w are respectively viscosity subsolution and supersolution of PPDE (1.1) and $v(T,\,\cdot\,) \le w(T,\,\cdot\,)$, then $v \le w$ on Θ. Then, the PPDE (1.1) admits a unique viscosity solution u given by the limit*

$$u(t,\omega) = \lim_{h\to 0} u^h(t,\omega), \tag{3.5}$$

locally uniformly in $(t,\omega) \in \Theta$.

We refer to Theorem 4.2 of [RTZ17] for sufficient conditions on PPDE coefficients for the comparison principle of viscosity solutions to be satisfied, see also Section 5.1. Convergence rates of $O(h^{1/10})$ and $O(h)$ have been derived for (3.5) respectively in [FTW11] for PDEs of Hamilton-Jacobi-Bellman type and in [ZZ14] for PPDEs under smoothness conditions, while the convergence rate of PPDE solutions remains unknown without smoothness conditions.

4 Deep learning approximation

In [RT17], the numerical estimation of the conditional expectation in (3.3) has been implemented using regression as in [GLW05], which requires to choose a basis for the functional space to be projected on. For example, assuming that $F(t,\omega,u,z,\gamma)$ takes the form

$$F(t,\omega,u,z,\gamma) = \widetilde{F}\left(t,\omega_t,\int_0^t \omega_s ds,u,z,\gamma\right)$$

and that g in (1.1) takes the form

$$g(\omega) = \widetilde{g}\left(\omega_T,\int_0^T \omega_s ds\right),$$

it can be reasonably guessed that the actual solution u will be of the form

$$u(t,\omega) = \widetilde{u}\left(t,\omega_t,\int_0^t \omega_s ds\right),$$

which motivates the choice of basis

$$\left(1,\omega_t,\int_0^t \omega_s ds,\omega_t^2,\left(\int_0^t \omega_s ds\right)^2,\omega_t\int_0^t \omega_s ds\right),$$

that uses second order polynomials. However, when g is not expressed in such form, e.g. when

$$g(\omega) = \widetilde{g}\left(\omega_T,\sup_{s\in[0,T]}\omega_s\right),$$

it is less clear how the actual solution u will look like, making it difficult to pick a suitable basis for the projection. Here, we overcome this difficulty by an alternative deep learning approach that does not rely on a specific form for the actual solution u, and has been previously applied with success to various high-dimensional problems, see e.g. [HJE18], [HPW20], [BBC+21].

Given $\rho : \mathbb{R} \to \mathbb{R}$ an activation function such as $\rho_{\mathrm{ReLU}}(x) := \max(0,x)$, $\rho_{\tanh}(x) := \tanh(x)$, or $\rho_{\mathrm{Id}}(x) := x$, we define the set of layer functions $\mathbb{L}^{\rho}_{d_1,d_2}$ by

$$\mathbb{L}^{\rho}_{d_1,d_2} := \left\{L : \mathbb{R}^{d_1} \to \mathbb{R}^{d_2} \ : \ L(x) = \rho(Wx+b),\ x\in\mathbb{R}^{d_1},\ W\in\mathbb{R}^{d_2\times d_1},\ b\in\mathbb{R}^{d_2}\right\},$$

where $d_1 \geq 1$ is the input dimension, $d_2 \geq 1$ is the output dimension, and the activation function ρ is applied component-wise to $Wx+b$. Then, we denote by

$$\mathbb{NN}^{\rho,l,m}_{d_0,d_1} := \left\{L_l \circ \cdots \circ L_0 : \mathbb{R}^{d_0} \to \mathbb{R}^{d_1} \ : \ L_0 \in \mathbb{L}^{\rho}_{d_0,m}, L_l \in \mathbb{L}^{\rho_{\mathrm{Id}}}_{m,d_1}, L_i \in \mathbb{L}^{\rho}_{m,m}, 1 \leq i < l\right\}$$

the set of feed-forward neural networks with one output layer, $l \geq 1$ hidden layers each containing $m \geq 1$ neurons, and the activation functions of the output and hidden layers being respectively the identity function ρ_{Id} and ρ. Any $L_l \circ \cdots \circ L_0 \in \mathbb{NN}_{d_0,d_1}^{\rho,l,m}$ is fully determined by the sequence

$$\theta := \big(W_0, b_0, W_1, b_1, \ldots, W_{l-1}, b_{l-1}, W_l, b_l\big)$$

of $((d_0+1)m + (l-1)(m+1)m + (m+1)d_1)$ parameters, such that

$$L_i(x) = \rho(W_i x + b_i), \qquad i = 0, 1, \ldots, l.$$

Building on (3.1)-(3.2), we let X^π denote the discretization

$$X_0^\pi = x_0, \qquad X_{i+1}^\pi = \begin{pmatrix} X_i^\pi \\ X_h^{(ih, \overline{X}_i^\pi)}(1) - x_0 + X_i^\pi(i) \end{pmatrix}, \quad i = 0, 1, \ldots, N-1, \tag{4.1}$$

where $X_i^\pi(k) \in \mathbb{R}^d$ is the *k-th* entry of the zero-based array $X_i^\pi \in \mathbb{R}^{(i+1)d}$ for $0 \leq k \leq i \leq N$. The same notation is used for $X_h^{(ih,\overline{X}_i^\pi)}$. Next, we introduce our deep learning scheme for the approximation of (3.4). In order not to overload the notation, we use the same notation θ for the parameters of the three neural networks used in the algorithm.

Algorithm 4.1 *i) Fix $(d, N, l, (m_i)_{0\leq i<N})$, the activation function ρ, initialize $\widehat{\mathcal{V}}_N : \mathbb{R}^{(N+1)d} \to \mathbb{R}$ by $\widehat{\mathcal{V}}_N(x) := g(\overline{x})$, $x \in \mathbb{R}^{(N+1)d}$.*

ii) For $i = N-1, \ldots, 0$, given $\widehat{\mathcal{V}}_{i+1} : \mathbb{R}^{(i+2)d} \to \mathbb{R}$,

a) Initialize the neural networks

$$(\mathcal{Y}_i(\cdot\,;\theta), \mathcal{Z}_i(\cdot\,;\theta), \gamma_i(\cdot\,;\theta)) \in \mathbb{NN}_{(i+1)d,1}^{\rho,l,m_i} \times \mathbb{NN}_{(i+1)d,d}^{\rho,l,m_i} \times \mathbb{NN}_{(i+1)d,d(d+1)/2}^{\rho,l,m_i},$$

b) Compute the mean square error function

$$\begin{aligned} E_i(\theta) \;:=\; & \mathbb{E}\big[\big|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_0^h - \mathcal{Y}_i\big(X_i^\pi;\theta\big)\big|^2 + \big\|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_1^h - \mathcal{Z}_i\big(X_i^\pi;\theta\big)\big\|_d^2 \\ & + \big\|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_2^h - \mathrm{Sym}\big(\gamma_i\big(X_i^\pi;\theta\big)\big)\big\|_{d\times d}^2\big], \end{aligned} \tag{4.2}$$

where for any sequence $(a_1, \ldots, a_{d(d+1)/2}) \in \mathbb{R}^{d(d+1)/2}$ we let

$$\mathrm{Sym}\big((a_1, \ldots, a_{d(d+1)/2})^\top\big) := \begin{pmatrix} 2a_{d(d-1)/2+1} & a_{d(d+1)/2-1} & \cdots & a_2 & a_1 \\ a_{d(d+1)/2-1} & \ddots & \ddots & \ddots & a_3 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ a_2 & \ddots & \ddots & \ddots & a_{d(d-1)/2} \\ a_1 & a_3 & \cdots & a_{d(d-1)/2} & 2a_{d(d+1)/2} \end{pmatrix},$$

c) Assuming the existence of $\min_\theta E_i(\theta)$*, we choose*

$$\theta_i^* \in \arg\min_\theta E_i(\theta).$$

d) Update $\big(\widehat{\mathcal{Y}}_i(\cdot), \widehat{\mathcal{Z}}_i(\cdot), \widehat{\gamma}_i(\cdot)\big) = (\mathcal{Y}_i(\cdot\,;\theta_i^*), \mathcal{Z}_i(\cdot\,;\theta_i^*), \gamma_i(\cdot\,;\theta_i^*))$ *and* $\widehat{\mathcal{V}}_i : \mathbb{R}^{(i+1)d} \to \mathbb{R}$ *by*

$$\widehat{\mathcal{V}}_i(x) := \widehat{\mathcal{Y}}_i(x) + hF\big(ih, \overline{x}, \widehat{\mathcal{Y}}_i(x), \widehat{\mathcal{Z}}_i(x), \mathrm{Sym}\,(\widehat{\gamma}_i(x))\,\big). \tag{4.3}$$

We note that minimizing the error function $E_i(\theta)$ is equivalent to minimizing the quantity

$$\begin{aligned}\varepsilon_i^{l,m,\theta} \;:=\; & \mathbb{E}\big[\big|\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_0^h\big] - \mathcal{Y}_i\left(X_i^\pi;\theta\right)\big|^2 + \big\|\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_1^h\big] - \mathcal{Z}_i\left(X_i^\pi;\theta\right)\big\|_d^2 \\ & + \big\|\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h\big] - \mathrm{Sym}\left(\gamma_i\left(X_i^\pi;\theta\right)\right)\big\|_{d\times d}^2\big],\end{aligned} \tag{4.4}$$

from the relationship

$$\begin{aligned}E_i(\theta) &= \mathbb{E}\big[\big|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_0^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_0^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_0^h\big] - \mathcal{Y}_i\big(X_i^\pi;\theta\big)\big|^2 \\ &\quad + \big\|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_1^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_1^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_1^h\big] - \mathcal{Z}_i\big(X_i^\pi;\theta\big)\big\|_d^2 \\ &\quad + \big\|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_2^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_2^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_2^h\big] - \mathrm{Sym}\big(\gamma_i\big(X_i^\pi;\theta\big)\big)\big\|_{d\times d}^2\big] \\ &= \mathbb{E}\big[\big|\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_0^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_0^h\big]\big|^2 + \big\|\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_1^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_1^h\big]\big\|_d^2 \\ &\quad + \big\|\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_2^h - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right)H_2^h\big]\big\|_{d\times d}^2\big] + \varepsilon_i^{l,m,\theta},\end{aligned}$$

where in the second equality, we have used the fact that for any square-integrable $\mathcal{F}_i$-measurable random variable Y,

$$\mathbb{E}[(\mathbb{E}_i[X] - Y)(X - \mathbb{E}_i[X])] = \mathbb{E}[(\mathbb{E}_i[X] - Y)\mathbb{E}_i[X - \mathbb{E}_i[X]]] = 0.$$

The next result provides an error bound of the Algorithm 4.1, where K is the Lipschitz constant in Assumption 1-(v) and $\widetilde{K}$ is the bounding constant in Assumption 1-(ii)

Proposition 4.2 *Using* (3.4) *and the notation of Algorithm 4.1, and assuming Assumptions 1-*(ii) *and 1-*(v)*, we have*

$$\max_{i=0,\ldots,N-1} \mathbb{E}\big[\big|\widehat{\mathcal{V}}_i\left(X_i^\pi\right) - u^h\big(ih, \overline{X}_{ih}^\pi\big)\big|^2\big] \le M\frac{L^N - 1}{L - 1}\varepsilon^{l,m},$$

where we let $L := 32(1 + K^2h^2 + K^2\widetilde{K}^2hd + K^2\widetilde{K}^4d(d+1))$*,* $M := 32(1 + K^2h^2)$*, and*

$$\varepsilon^{l,m} := \sum_{i=0}^{N-1} \varepsilon_i^{l,m,\theta_i^*}. \tag{4.5}$$

The proof of Proposition 4.2, see Appendix A, uses only the Lipschitz continuity of F and the boundedness of σ^{-1} in Assumption 1 , while the rest of the conditions in Assumption 1 are required in Theorem 3.1 as in [FTW11] and [RT17]. Next, we recall the following universal approximation theorem.

Theorem 4.3 *(Theorem 1 in [Hor91]). Fix $l \geq 1$, if the activation function ρ is unbounded and nonconstant, then for any finite measure μ the set $\bigcup_{m=1}^{\infty} \mathbb{NN}_{d_0,1}^{\rho,l,m}$ is dense in $L^q(\mu)$ for all $q \geq 1$.*

The next corollary shows that the neural network approximation can be made arbitrarily close to the PPDE solution $u\left(0,(x_0)_{s\in[0,T]}\right)$.

Theorem 4.4 *Under the assumptions of Theorems 3.1 and 4.3, assume additionally that the activation function ρ is Lipschitz. Then, for any $\varepsilon > 0$ there exists $(m_i)_{0\leq i<N}$ and $(\theta_i^*)_{0\leq i<N}$ such that $(\widehat{\mathcal{V}}_i)_{0\leq i\leq N}$ constructed from $(m_i)_{0\leq i<N}$ and $(\theta_i^*)_{0\leq i<N}$ in (4.3) satisfies*

$$\left|u(0,(x_0)_{s\in[0,T]}) - \widehat{\mathcal{V}}_0(x_0)\right| < \varepsilon.$$

Proof. Let $\varepsilon > 0$. By Theorem 3.1, we can find $h > 0$ small enough such that

$$\left|u\left(0,(x_0)_{s\in[0,T]}\right) - u^h\left(0,(x_0)_{s\in[0,T]}\right)\right| < \frac{\varepsilon}{2}.$$

First, we note that by Proposition 4.2, the proof is complete by the triangle inequality if we can choose $(m_i)_{0\leq i<N}$ and $(\theta_i^*)_{0\leq i<N}$ such that $\varepsilon^{l,m}$ defined by (4.4) and (4.5) satisfies

$$\begin{aligned}\varepsilon^{l,m} = \sum_{i=0}^{N-1} \mathbb{E}\Big[&\left|\mathbb{E}_i\left[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_0^h\right] - \mathcal{Y}_i\left(X_i^\pi;\theta_i^*\right)\right|^2 + \left\|\mathbb{E}_i\left[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_1^h\right] - \mathcal{Z}_i\left(X_i^\pi;\theta_i^*\right)\right\|_d^2 \\ &+ \left\|\mathbb{E}_i\left[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h\right] - \mathrm{Sym}\left(\gamma_i\left(X_i^\pi;\theta_i^*\right)\right)\right\|_{d\times d}^2\Big] \;<\; \frac{L-1}{2M(L^N-1)}\varepsilon, \end{aligned} \tag{4.6}$$

Next, we note that (4.6) holds if we show that

$$\mathbb{E}\left[\left\|\mathbb{E}_i\left[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h\right] - \mathrm{Sym}\left(\gamma_i\left(X_i^\pi;\theta_i^*\right)\right)\right\|_{d\times d}^2\right] < \frac{L-1}{6NM(L^N-1)}\varepsilon,$$

$i = 0,\ldots,N-1$, as the argument for the other terms is similar. For this, we rely on the universal approximation Theorem 4.3, which requires us to show that $\mathbb{E}_i\left[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h\right]$, as a function of X_i on $\mathbb{R}^{(i+1)d} \to \mathbb{S}^d$, is in $L^2(\mu)$, where μ is the joint distribution of X_i^π. By the Lipschitz condition on $b(t,\omega)$ and $\sigma(t,\omega)$ in Assumption 1 we have

$$\mathbb{E}\left[\max_{0\leq k\leq i}\|X_i^\pi(k)\|_d^q\right] < \infty \qquad \text{and} \qquad \mathbb{E}\left[\|\varphi(X_i^\pi)\|_k^q\right] < \infty, \tag{4.7}$$

for any Lipschitz continuous function $\varphi : \mathbb{R}^{(i+1)d} \to \mathbb{R}^k$ and all $q \geq 1$, $0 \leq i \leq N$, see Appendix B. Hence, by Assumptions 1-(iii), 1-(v), and Hölder's inequality, we have $\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_2\big] = \mathbb{E}_i\big[g\big(\overline{X}^\pi_{(i+1)h}\big)H^h_2\big] \in L^2(\mu)$ at the level $i = N-1$. For $0 \leq i < N$, using Assumptions 1-(ii) and 1-(v), we have

$$
\begin{aligned}
&\mathbb{E}\big[\big\|\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_2\big]\big\|^2_{d\times d}\big] \\
&\leq \mathbb{E}\big[\|H^h_2\|^2_{d\times d} \times \big|\widehat{\mathcal{Y}}_{i+1}(X^\pi_{i+1}) + hF\big((i+1)h, \overline{X}^\pi_{(i+1)h}, \widehat{\mathcal{Y}}_{i+1}(X^\pi_{i+1}), \widehat{\mathcal{Z}}_{i+1}(X^\pi_{i+1}), \mathrm{Sym}\left(\widehat{\gamma}_{i+1}(X^\pi_{i+1})\right)\big)\big|^2\big] \\
&\leq 6^3\Big(\mathbb{E}\big[\|H^h_2\|^4_{d\times d}\big]\mathbb{E}\big[|\widehat{\mathcal{Y}}_{i+1}(X^\pi_{i+1})|^4 + h^4|F((i+1)h, (0)_{0\leq s\leq T}, 0, 0, 0)|^4 \\
&\qquad + h^4K^4\big(\|\overline{X}^\pi_{(i+1)h}\|^4 + |\widehat{\mathcal{Y}}_{i+1}(X^\pi_{i+1})|^4 + \|\widehat{\mathcal{Z}}_{i+1}(X^\pi_{i+1})\|^4_d + \|\mathrm{Sym}\big(\widehat{\gamma}_{i+1}(X^\pi_{i+1})\big)\|^4_{d\times d}\big)\big]\Big)^{1/2} \\
&< \infty,
\end{aligned}
$$

which shows that $\mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_2\big] \in L^2(\mu)$. In the last inequality we have used (4.7), the fact that $\varphi \in \mathbb{NN}^{\rho,l,m}_{d_0,d_1}$ is Lipschitz when the activation function ρ is Lipschitz, and $\mathbb{E}\left[\|H^h_2\|^4_{d\times d}\right] < \infty$ with $\|\overline{X}^\pi_{ih}\| := \max_{0\leq k\leq i}\|X^\pi_i(k)\|_d$. □

Using additionally that $u \in \mathrm{BUC}(\Theta)$ is uniformly continuous, Proposition 4.2 and Theorem 4.4 can be extended from $(0, (x_0)_{s\in[0,T]})$ to any $(t,\omega) \in \Theta$ by changing (4.1) to start from $X^\pi_{kh} = (\omega^\pi_{s\wedge kh})_{s\in[0,T]}$, where ω^π is the linear interpolation of the discretization of ω.

Implementation

The optimization in (4.2) is implemented using Monte Carlo simulation and the Adam gradient descent algorithm, see [KB14]. Precisely, fix the batch size O and the training steps P, let $\big(X^{\pi,j}_{i+1}\big)_{1\leq j\leq O}$ be an i.i.d. sample of $(i+2)d$-dimensional random vector with batch size O generated by (4.1), and let

$$
\begin{aligned}
L^O_i(\theta) := \frac{1}{O}\sum_{j=1}^{O}\Big(&\big|\widehat{\mathcal{V}}_{i+1}\big(X^{\pi,j}_{i+1}\big)H^h_0 - \mathcal{Y}_i\big(X^{\pi,j}_i;\theta\big)\big|^2 + \big\|\widehat{\mathcal{V}}_{i+1}\big(X^{\pi,j}_{i+1}\big)H^h_1 - \mathcal{Z}_i\big(X^{\pi,j}_i;\theta\big)\big\|^2_d \\
&+ \big\|\widehat{\mathcal{V}}_{i+1}\big(X^{\pi,j}_{i+1}\big)H^h_2 - \mathrm{Sym}\big(\gamma_i\big(X^{\pi,j}_i;\theta\big)\big)\big\|^2_{d\times d}\Big).
\end{aligned}
$$

Since in (4.2) the error term of $\mathcal{Y}$ becomes negligible in the high-dimensional case, we use the weighted error function of

$$
\begin{aligned}
\mathbb{E}\big[\big|\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_0 - \mathcal{Y}_i\big(X^\pi_i;\theta\big)\big|^2 + \tfrac{1}{d}\big\|\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_1 - \mathcal{Z}_i\big(X^\pi_i;\theta\big)\big\|^2_d \\
+\tfrac{1}{d^2}\big\|\widehat{\mathcal{V}}_{i+1}\big(X^\pi_{i+1}\big)H^h_2 - \mathrm{Sym}\big(\gamma_i\big(X^\pi_i;\theta\big)\big)\big\|^2_{d\times d}\big].
\end{aligned}
$$

Then, we initialize the parameter θ_0 using Xavier initialization, see [GB10], and update it using the following rule

$$\begin{cases} v_p &= \beta_1 v_{p-1} + (1-\beta_1)\dfrac{\partial L_i^O}{\partial \theta}(\theta_{p-1}) \\ w_p &= \beta_2 w_{p-1} + (1-\beta_2)\left(\dfrac{\partial L_i^O}{\partial \theta}(\theta_{p-1})\right)^2 \\ \theta_p &= \theta_{p-1} - \eta_p \left(\dfrac{v_p}{1-\beta_1}\right) \Bigg/ \left(\varepsilon_{\text{Adam}} + \sqrt{\dfrac{w_p}{1-\beta_1}}\right), \end{cases}$$

where $1 \le p \le P$, $(\eta_p)_{1\le p\le P} \in \mathbb{R}^P$ is the learning rate, $(\varepsilon_{\text{Adam}}, \beta_1, \beta_2) \in \mathbb{R}^3$ are the parameters of the Adam algorithm, and (v_0, w_0) is initialized at $(0,0)$. Empirically we have $\theta_P \approx \theta^*$ when O and P are large enough, see e.g. [KB14]. In addition, we use the batch normalization technique, see [IS15], to stabilize the training process. Define $\text{BN}_{\alpha,\zeta,\varepsilon_{\text{BN}}}$ a transformation over a set of d_1-dimensional $\left(x_j^{(i)}\right)_{1\le i\le O, 1\le j\le d_1}$ with batch size O by

$$\text{BN}_{\alpha,\zeta,\varepsilon_{\text{BN}}}\left(x^{(i)}\right) = \left(\zeta_j + \alpha_j\left(x_j^{(i)} - \mu_j\right)\Big/\sqrt{\sigma_j^2 + \varepsilon_{\text{BN}}}\ \right)_{1\le j\le d_1}, \tag{4.8}$$

where $\alpha, \zeta \in \mathbb{R}^{d_1}$, $\varepsilon_{\text{BN}} \in \mathbb{R}$, and

$$\mu_j = \frac{1}{O}\sum_{i=1}^{O} x_j^{(i)} \quad \text{and} \quad \sigma_j^2 = \frac{1}{O}\sum_{i=1}^{O}(x_j^{(i)} - \mu_j)^2.$$

Fix $\varepsilon_{\text{BN}} \in \mathbb{R}$, a neural network $\varphi(\cdot\,;\theta) \in \mathbb{NN}_{d_0,d_1}^{\rho,l,m}$ is modified such that each of the layer functions $L_i \in \mathbb{L}_{d_2,d_3}^{\rho}$ is changed to $L_i \in \mathbb{L}_{d_2,d_3}^{\rho,\text{BN}}$, where

$$\mathbb{L}_{d_2,d_3}^{\rho,\text{BN}} := \left\{ L : \mathbb{R}^{d_2} \to \mathbb{R}^{d_3} \ : \ L(x) = \text{BN}_{\alpha,\zeta,\varepsilon_{\text{BN}}}\left(\rho(Wx+b)\right), W \in \mathbb{R}^{d_3\times d_2}, \alpha, b, \zeta \in \mathbb{R}^{d_3} \right\},$$

and a transformation from $x \in \mathbb{R}^{d_0}$ to $\text{BN}_{\gamma_{-1},\beta_{-1},\varepsilon_{\text{BN}}}(x)$, parameterized by $\gamma_{-1}, \beta_{-1} \in \mathbb{R}^{d_0}$, is added before passing to the first layer. Then, the neural network parameter θ is changed to the sequence

$$\Theta_{\text{BN}} = \left(\gamma_{-1}, \beta_{-1}, W_0, b_0, \gamma_0, \beta_0, W_1, b_1, \gamma_1, \beta_1, \ldots, W_{l-1}, b_{l-1}, \gamma_{l-1}, \beta_{l-1}, W_l, b_l, \gamma_l, \beta_l\right)$$

of parameters, such that

$$L_i(x) = \text{BN}_{\gamma_i,\beta_i,\varepsilon_{\text{BN}}}\left(\rho(W_i x + b_i)\right), \qquad i = 0, 1, \ldots, l.$$

In Section 5 we provide three examples of implementation of the numerical scheme of Proposition 4.2. In our numerical examples we use the activation function $\rho = \rho_{\text{ReLU}}$, the Adam parameters $(\beta_1, \beta_2, \varepsilon_{\text{Adam}}) = (0.9, 0.999, 10^{-8})$, the batch normalization parameter $\varepsilon_{\text{BN}} = 10^{-6}$,

and the learning rate

$$\eta_p = \begin{cases} 10^{-1}, & 1 \le p < 2P/3, \\ 10^{-2}, & 2P/3 \le p < 5P/6, \\ 10^{-3}, & 5P/6 \le p < P. \end{cases}$$

The values of P have been set so that runtimes are comparable when $d = 1$, and reused for simulations with $d > 1$.

As in [AA13], for improved convergence our numerical implementation will also use the modification

$$\begin{cases} Y_h\phi(t,\omega) := \mathbb{E}\left[\phi\left(t+h, \omega \otimes_t \overline{X}_h^{(t,\omega)}\right) H_0^h \,\middle|\, \mathcal{F}_t\right], \\ Z_h\phi(t,\omega) := \mathbb{E}\left[\left(\phi\left(t+h, \omega \otimes_t \overline{X}_h^{(t,\omega)}\right) - Y_h\phi(t,\omega)\right) H_1^h \,\middle|\, \mathcal{F}_t\right], \\ \Gamma_h\phi(t,\omega) := \mathbb{E}\left[\left(\phi\left(t+h, \omega \otimes_t \overline{X}_h^{(t,\omega)}\right) - Y_h\phi(t,\omega) - Z_h\phi(t,\omega)\cdot W_h\right) H_2^h \,\middle|\, \mathcal{F}_t\right], \end{cases}$$

of (3.4) where the operator $\mathbb{T}^{t,\omega}$ in (3.3) is replaced by

$$\mathbb{T}^{t,\omega}\left[u^h(t+h,\,\cdot\,)\right] = Y_h\phi(t,\omega) + hF\left(\,\cdot\,, Y_h\phi, Z_h\phi, \Gamma_h\phi\right)(t,\omega), \tag{4.9}$$

and the error function $E_i(\theta)$ in (4.2) is replaced with

$$\begin{aligned} E_i(\theta) = \mathbb{E}\big[\big|\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big)H_0^h - \mathcal{Y}_i\big(X_i^\pi;\theta\big)\big|^2 + \big\|\big(\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big) - \mathcal{Y}_i\big(X_i^\pi;\theta\big)\big)H_1^h - \mathcal{Z}_i\big(X_i^\pi;\theta\big)\big\|_d^2 \\ + \big\|\big(\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big) - \mathcal{Y}_i\big(X_i^\pi;\theta\big) - \mathcal{Z}_i\big(X_i^\pi;\theta\big)\cdot W_h\big)H_2^h - \gamma_i\big(X_i^\pi;\theta\big)\big\|_{d\times d}^2\big]. \end{aligned} \tag{4.10}$$

Although the following examples do not satisfy all conditions stated in Assumption 1, we will use them as in e.g. [RT17] to assess the performance of Algorithm 4.1.

5 Numerical examples

Our examples are run with $T = 0.1$, for which our algorithm performs optimally and can improve the results of [RT17], [SVŠS20] and [SZ21] for the zero-sum game, of [SVŠS20, SZ21] in dimension $d = 10$ for Asian options, and of [RT17, SVŠS20, SZ21] respectively in dimensions $d = 100$ and $d = 10$ for barrier options. Due to the different nature and implementation of the algorithms, runtime comparison is used for reference, but is not fully reliable.

5.1 Path-dependent two-person zero-sum game

In this section we consider the higher dimensional extension

$$u(0,(x_0)_{s\in[0,T]}) = \inf_{\mu_t\in[\underline{\mu},\overline{\mu}]} \sup_{a_t\in[\underline{a},\overline{a}]} \mathbb{E}\left[g\left(X_T^{\mu,a}, \int_0^T X_s^{\mu,a} ds\right) + \int_0^T f\left(t, X_t^{\mu,a}, \int_0^t X_s^{\mu,a} ds\right) dt\right],$$

of the path-dependent two-person zero-sum game in (5.1) of [RT17], where $(X_t)_{0\le t\le T} = (X_t^1,\dots,X_t^d)_{0\le t\le T}$ follows the SDE

$$\begin{cases} dX_t^{\mu,a} = \mu_t \mathbf{1}_d dt + \sqrt{a_t}\, \boldsymbol{I}_d dB_t, \\ X_0 = x_0 \in \mathbb{R}^d, \end{cases} \tag{5.1}$$

where $\underline{\mu}$, $\overline{\mu}$, $\underline{a}$, $\overline{a} \in \mathbb{R}$, and $(B_t)_{0\le t\le T} = (B_t^1,\dots,B_t^d)_{0\le t\le T}$ is a d-dimensional Brownian motion. The solution of this control problem is given by the solution $u : [0,T]\times C([0,T];\mathbb{R}^d) \to \mathbb{R}$ of the following PPDE, evaluated at $(0,(x_0)_{s\in[0,T]})$:

$$\begin{cases} \partial_t u(t,\omega) + \min\limits_{\mu\in[\underline{\mu},\overline{\mu}]} \mu\big(\mathbf{1}_d\cdot\partial_\omega u(t,\omega)\big) + \dfrac{1}{2}\max\limits_{a\in[\underline{a},\overline{a}]}\left(a\,\mathrm{tr}\left(\partial^2_{\omega\omega}u(t,\omega)\right)\right) + f\left(t,\omega_t,\displaystyle\int_0^t \omega_s ds\right) = 0, \\ u(T,\omega) = g\left(\omega_T, \displaystyle\int_0^T \omega_s ds\right). \end{cases}$$

For the purpose of our deep algorithm, we rewrite the above PPDE as

$$\partial_t u + \frac{\underline{a}}{2}\,\mathrm{tr}\left(\partial^2_{\omega\omega}u\right) + \min_{\mu\in[\underline{\mu},\overline{\mu}]} \mu\left(\mathbf{1}_d\cdot\partial_\omega u\right) + \frac{1}{2}\max_{a\in[\underline{a},\overline{a}]} a\,\mathrm{tr}\left(\partial^2_{\omega\omega}u\right) + f\left(t,\omega_t,\int_0^t\omega_s ds\right) - \frac{\underline{a}}{2}\,\mathrm{tr}\left(\partial^2_{\omega\omega}u\right) = 0, \tag{5.2}$$

with

$$F(t,\omega,u,z,\gamma) = \frac{1}{\sqrt{\underline{a}}}\min_{\mu\in[\underline{\mu},\overline{\mu}]}\mu\left(\mathbf{1}_d\cdot z\right) + \frac{1}{2\underline{a}}\max_{a\in[\underline{a},\overline{a}]}\left(a\,\mathrm{tr}\,\gamma\right) + f\left(t,\omega_t,\int_0^t\omega_s ds\right) - \frac{1}{2}\,\mathrm{tr}\,\gamma.$$

Denoting by $x = (x^1,\dots,x^d)$ and $y = (y^1,\dots,y^d)$ we put $g(x,y) = \cos\left(\frac{1}{d}\sum_{i=1}^d (x^i+y^i)\right)$ and

$$\begin{aligned} f(t,x,y) = {} & \left(\frac{1}{d}\sum_{i=1}^d x^i + \overline{\mu}\right)\left(\sin\left(\frac{1}{d}\sum_{i=1}^d\left(x^i+y^i\right)\right)\right)^+ - \left(\frac{1}{d}\sum_{i=1}^d x^i + \underline{\mu}\right)\left(\sin\left(\frac{1}{d}\sum_{i=1}^d\left(x^i+y^i\right)\right)\right)^- \\ & + \frac{\underline{a}}{2d}\left(\cos\left(\frac{1}{d}\sum_{i=1}^d\left(x^i+y^i\right)\right)\right)^+ - \frac{\overline{a}}{2d}\left(\cos\left(\frac{1}{d}\sum_{i=1}^d\left(x^i+y^i\right)\right)\right)^-. \end{aligned}$$

Although this choice of $f(t,x,y)$ does not satisfy part (v) of Assumption 1, it makes the PPDE (5.2) explicitly solvable as $u(t,\omega) = \cos\left(\frac{1}{d}\sum_{i=1}^d\left(\omega_t^i + \int_0^t \omega_s^i ds\right)\right)$, which can be used to estimate the precision of Algorithm 4.1.

In Table 1, our PPDE algorithm is compared to [RT17] and [SZ21] with ten Monte Carlo runs, by taking $\underline{\mu} = -0.2$, $\overline{\mu} = 0.2$, $\underline{a} = 0.04$, $\overline{a} = 0.09$, $T = 0.1$, $x_0 = (0, \ldots, 0)$, and runtimes are measured in seconds.

Method	d	Regr./Deep	Mean	Stdev	Ref. value	Rel. L^1-error	Runtime (s)
Deep PPDE (4.10)	1	Deep	1.000805	9.61E-05	1.0	8.05E-04	62
Deep PPDE (4.2)	1	Deep	0.999331	1.52E-03	1.0	1.37E-03	61
[SZ21]	1	Deep	1.000852	1.12E-02	1.0	9.42E-03	26
[RT17] (4.9)	1	Regr.	0.999946	4.72E-05	1.0	5.47E-05	1
[RT17] (3.4)	1	Regr.	1.075509	2.59E-02	1.0	7.55E-02	1
Deep PPDE (4.10)	10	Deep	1.000914	2.19E-04	1.0	9.14E-04	63
Deep PPDE (4.2)	10	Deep	0.9939934	2.99E-03	1.0	6.01E-03	62
[SZ21]	10	Deep	0.9537241	1.63E-01	1.0	1.06E-01	517
[RT17] (4.9)	10	Regr.	1.000166	6.00E-06	1.0	1.66E-04	2
[RT17] (3.4)	10	Regr.	2.348812	4.31E-01	1.0	Diverges	2
Deep PPDE (4.10)	100	Deep	1.002474	5.01E-04	1.0	2.47E-03	83
Deep PPDE (4.2)	100	Deep	0.970783	1.89E-02	1.0	3.01E-02	81
[RT17] (4.9)	100	Regr.	26.12039	7.36E+00	1.0	Diverges	104
[RT17] (3.4)	100	Regr.	328.2853	8.81E+01	1.0	Diverges	102

Table 1: Comparison for two-person zero-sum game between *i*) PPDE with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 900$; *ii*) [RT17] with $O = 10000$ and $h = 0.01$; *iii*) [SZ21] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 1000$. The numerical simulations of [SZ21] are not presented in dimension $d = 100$ because they require more than the 12 GB RAM provided by Google Colab.

5.2 Asian options

The second example is the following pricing problem of Asian basket call option:

$$u(0, (x_0)_{s\in[0,T]}) = \mathbb{E}\left[e^{-r_0 T} \left(\frac{1}{Td} \sum_{i=1}^{d} \int_0^T X_s^i ds - K \right)^+ \right],$$

with strike price $K \in \mathbb{R}$ and interest rate $r_0 > 0$, where $(X_t)_{0\le t\le T} = (X_t^1, \ldots, X_t^d)_{0\le t\le T}$ is a d-dimensional asset price process following the geometric Brownian motions

$$X_t^i = X_0^i \mathrm{e}^{\sigma_i B_t^i + r_i t - \sigma_i^2 t/2}, \qquad t \in \mathbb{R}_+, \quad i = 1, \ldots, d, \tag{5.3}$$

where $x_0 \in \mathbb{R}^d$, and $r_1, \ldots, r_d, \sigma_1, \ldots, \sigma_d \in \mathbb{R}$. The solution of this pricing problem is given by evaluating at $(t, x) = (0, (x_0)_{s\in[0,T]})$ the solution $u : [0, T] \times C\left([0, T]; \mathbb{R}^d\right) \to \mathbb{R}$ of the

following PPDE:

$$\begin{cases} \partial_t u(t,\omega) + r(\omega_t) \cdot \partial_\omega u(t,\omega) + \dfrac{1}{2}\left(\sigma\sigma^\top(\omega_t) : \partial^2_{\omega\omega} u(t,\omega)\right) - r_0 u(t,\omega) = 0, \\ u(T,\omega) = \left(\dfrac{1}{Td}\displaystyle\sum_{i=1}^{d}\int_0^T \omega_s^i ds - K\right)^+, \end{cases}$$

where $r(\omega_t) = \left(r_1\omega_t^1, \ldots, r_d\omega_t^d\right)$ and $\sigma(\omega_t) = \mathrm{Diag}\left(\sigma_1\omega_t^1, \ldots, \sigma_d\omega_t^d\right)$. Here, $F(t,\omega,u,z,\gamma) = -r_0 u$ does not depend on z and γ, therefore the neural networks $\mathcal{Z}_i(\cdot\,;\theta)$ and $\gamma_i(\cdot\,;\theta)$ are not needed, which improves the efficiency of Algorithm 4.1.

When $d = 1$ we compare our deep PPDE algorithm with other deep PDE algorithms such as [HJE18] and [BBC+21]. For this, we write

$$u(t,(X_s)_{s\in[0,t]}) = g\left(t, \frac{1}{X_t}\left(\frac{1}{T}\int_0^t X_u du - K\right)\right),$$

where $g : [0,T]\times\mathbb{R}\to\mathbb{R}$ is the solution of the [RS95] PDE

$$\partial_t g + \left(\frac{1}{T} - rz\right)\frac{\partial g}{\partial x} + \frac{1}{2}\sigma^2 z^2 \partial^2_{zz} g = 0, \quad g(T,z) = z^+, \tag{5.4}$$

see e.g. Proposition 13.10 in [Pri22].

We use Monte Carlo simulations with $O = 1,000,000$ and $h = 0.01$ as the reference solution. to compare our PPDE algorithm with [RT17], [HJE18], [SVŠS20], [SZ21], and [BBC+21] under the setting of $r_0 = r_1 = \cdots = r_d = 0.01$, $\sigma_1 = \cdots = \sigma_d = 0.1$, $K = 0.7$, $T = 0.1$, $x_0 = (1,\ldots,1)$. The statistics of 10 independent runs are summarized in Table 2.

Method	d	Regr./Deep	Mean	Stdev	Ref. value	Rel. L^1-error	Runtime (s)
[HJE18]	1	Deep	0.3002467	2.31E-06	0.3002021	1.49E-04	32
[BBC+21]	1	Deep	0.3002827	4.21E-04	0.3002021	1.12E-03	20
[SVŠS20]	1	Deep	0.3002722	1.24E-03	0.3002021	3.51E-03	10
Deep PPDE (4.2)	1	Deep	0.3008159	1.29E-03	0.3002021	3.83E-03	31
[SZ21]	1	Deep	0.3002544	2.43E-03	0.3002021	6.01E-03	25
[RT17]	1	Regr.	0.3002768	2.23E-04	0.3002021	4.77E-04	1
Deep PPDE (4.2)	10	Deep	0.3010345	4.08E-04	0.3002024	2.77E-03	31
[SVŠS20]	10	Deep	0.3002251	2.11E-03	0.3002024	5.80E-03	411
[SZ21]	10	Deep	0.3040330	1.05E-02	0.3002024	2.97E-02	522
[RT17]	10	Regr.	0.3002137	6.07E-05	0.3002024	1.70E-04	2
Deep PPDE (4.2)	100	Deep	0.3006346	1.28E-04	0.3001993	1.45E-03	35
[RT17]	100	Regr.	0.3001923	2.65E-05	0.3001993	7.50E-05	23

Table 2: Comparison for Asian options between *i*) PPDE with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 900$; *ii*) [RT17] with $O = 10000$ and $h = 0.01$; *iii*) [SZ21] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and iterations 1000; *iv*) LSTM with path signature [SVŠS20] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 600$; *v*) [HJE18] with training parameters $m = d + 10$, $l = 2$, $O = 64$, $h = 0.01$, and $P = 4000$; *vi*) [BBC+21] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 600$. The numerical simulations of [SZ21] and [SVŠS20] are not presented in dimension $d = 100$ because they require more than the 12 GB RAM provided by Google Colab.

5.3 Barrier options

The third example is the following pricing problem of barrier basket call option:

$$u(0,(x_0)_{s\in[0,T]}) = \mathbb{E}\left[e^{-r_0 T}\mathbb{1}_{\left\{\max_{0\le s\le T}\left(\frac{1}{d}\sum_{i=1}^{d} X_s^i\right)<B\right\}}\left(\frac{1}{d}\sum_{i=1}^{d} X_T^i - K\right)^+\right],$$

where the strike price $K \in \mathbb{R}$, the barrier $B \in \mathbb{R}$, and $(X_t)_{0\le t\le T} = (X_t^1,\dots,X_t^d)_{0\le t\le T}$ is a d-dimensional stock processes that follows the geometric Brownian motions (5.3) The solution of this pricing problem is given by evaluating at $(t,x) = (0,(x_0)_{s\in[0,T]})$ the solution $u : [0,T]\times C\left([0,T];\mathbb{R}^d\right) \to \mathbb{R}$ of the following PPDE:

$$\begin{cases} \partial_t u(t,\omega) + r(\omega_t)\cdot\partial_\omega u(t,\omega) + \dfrac{1}{2}\left(\sigma\sigma^\top(\omega_t) : \partial^2_{\omega\omega} u(t,\omega)\right) - r_0 u(t,\omega) = 0, \\ u(T,\omega) = \mathbb{1}_{\left\{\max_{0\le s\le T}\left(\frac{1}{d}\sum_{i=1}^{d}\omega_s^i\right)<B\right\}}\left(\dfrac{1}{d}\sum_{i=1}^{d}\omega_T^i - K\right)^+. \end{cases}$$

As in Section 5.2, $F(t,\omega,u,z,\gamma) = -r_0 u$ does not depend on z and γ and the neural networks $\mathcal{Z}_i(\cdot\,;\theta)$, and $\gamma_i(\cdot\,;\theta)$ are not needed.

We use Monte Carlo simulations with $O = 1000000$ and $h = 0.01$ as the reference solution to compare our PPDE algorithm with [RT17], [SVŠS20], and [SZ21] under the setting of $r_0 = r_1 = \cdots = r_d = 0.01$, $\sigma_1 = \cdots = \sigma_d = 0.1$, $K = 0.7$, $B = 1.2$, $T = 0.1$, $x_0 = (1, \ldots, 1)$. The statistics of 10 independent runs are summarized in Table 3.

Method	d	Regr./Deep	Mean	Stdev	Ref. value	Rel. L^1-error	Runtime (s)
[SVŠS20]	1	Deep	0.3009402	2.18E-03	0.3007008	5.75E-03	8
Deep PPDE (4.2)	1	Deep	0.3019161	1.97E-03	0.3007008	5.92E-03	31
[SZ21]	1	Deep	0.3019159	2.24E-03	0.3007008	6.98E-03	26
[RT17]	1	Regr.	0.3006738	2.81E-04	0.3007008	7.72E-04	1
Deep PPDE (4.2)	10	Deep	0.3017532	5.44E-04	0.3006973	3.51E-03	31
[SVŠS20]	10	Deep	0.301107	3.15E-03	0.3006973	7.35E-03	225
[SZ21]	10	Deep	0.3030515	1.03E-02	0.3006973	2.71E-02	519
[RT17]	10	Regr.	0.3007255	1.34E-04	0.3006973	3.56E-04	2
Deep PPDE (4.2)	100	Deep	0.3016375	1.98E-04	0.3007003	3.12E-03	35
[RT17]	100	Regr.	0.3035602	3.74E-03	0.3007003	1.05E-02	23

Table 3: Comparison for barrier options between *i)* PPDE with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 900$; *ii)* [RT17] with $O = 10000$ and $h = 0.01$; *iii)* [SZ21] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 1000$; *iv)* LSTM with path signature [SVŠS20] with training parameters $m = d + 10$, $l = 2$, $O = 256$, $h = 0.01$, and $P = 600$. The numerical simulations of [SZ21] and [SVŠS20] are not presented in dimension $d = 100$ because they require more than the 12 GB RAM provided by Google Colab.

A Appendix

Proof of Proposition 4.2. Let $\delta_i := \widehat{\mathcal{V}}_i\left(X_i^\pi\right) - u^h\left(ih, \overline{X}_{ih}^\pi\right)$, $i = 0, \ldots, N-1$. By (3.4), (4.3), Assumptions 1-(ii), 1-(v), and the conditional Hölder inequality, we have

$$
\begin{aligned}
&\mathbb{E}\left[|\delta_i|^2\right] = \mathbb{E}\big[\big|\widehat{\mathcal{Y}}_i\left(X_i^\pi\right) + hF\big(ih, \overline{X}_{ih}^\pi, \widehat{\mathcal{Y}}_i\left(X_i^\pi\right), \widehat{\mathcal{Z}}_i\left(X_i^\pi\right), \mathrm{Sym}\left(\widehat{\gamma}_i\left(X_i^\pi\right)\right)\big) - \mathbb{E}_i\left[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big)\right] \\
&+ hF\big(ih, \overline{X}_{ih}^\pi, \mathbb{E}_i\big[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_0^h\big], \mathbb{E}_i\big[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_1^h\big], \mathbb{E}_i\big[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi H_2^h\big)\big]\big)\big|^2\big] \\
&\leq 16(1+K^2h^2)\mathbb{E}\big[\big|\widehat{\mathcal{Y}}_i\left(X_i^\pi\right) - \mathbb{E}_i\left[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_0^h\right]\big|^2\big] \\
&\quad + 16K^2h^2\mathbb{E}\big[\big\|\widehat{\mathcal{Z}}_i\big(X_i^\pi\big) - \mathbb{E}_i\big[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_1^h\big]\big\|_d^2 \\
&\quad + \big\|\mathrm{Sym}\big(\widehat{\gamma}_i\left(X_i^\pi\right)\big) - \mathbb{E}_i\big[u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_2^h\big]\big\|_{d\times d}^2\big] \\
&= 16(1+K^2h^2)\mathbb{E}\big[\big|\widehat{\mathcal{Y}}_i\left(X_i^\pi\right) - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\big(X_{i+1}^\pi\big) H_0^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_0^h - u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_0^h\big]\big|^2\big] \\
&\quad + 16K^2h^2\mathbb{E}\big[\big\|\widehat{\mathcal{Z}}_i\left(X_i^\pi\right) - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_1^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_1^h - u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_1^h\big]\big\|_d^2 \\
&\quad + \big\|\mathrm{Sym}\left(\widehat{\gamma}_i\left(X_i^\pi\right)\right) - \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h\big] + \mathbb{E}_i\big[\widehat{\mathcal{V}}_{i+1}\left(X_{i+1}^\pi\right) H_2^h - u^h\big((i+1)h, \overline{X}_{(i+1)h}^\pi\big) H_2^h\big]\big\|_{d\times d}^2\big]
\end{aligned}
$$

$$
\begin{aligned}
&\le 32(1+K^2h^2)\varepsilon_i^{l,m,\theta^*} + 32\mathbb{E}\big[\mathbb{E}_i[|\delta_{i+1}|^2]\mathbb{E}_i\big[(1+K^2h^2)\left|H_0^h\right|^2 + K^2h^2\left\|H_1^h\right\|_d^2 + K^2h^2\left\|H_2^h\right\|_{d\times d}^2\big]\big]\\
&\le 32(1+K^2h^2)\varepsilon_i^{l,m,\theta^*} + 32\left(1+K^2h^2+K^2\widetilde{K}^2hd+K^2\widetilde{K}^4d(d+1)\right)\mathbb{E}\left[|\delta_{i+1}|^2\right]\\
&= M\varepsilon_i^{l,m,\theta^*} + L\mathbb{E}\left[|\delta_{i+1}|^2\right], \quad i=0,\dots,N-2.
\end{aligned}
$$

Using backward induction and the fact that $\widehat{\mathcal{V}}_N(x) = u^h(t_N,\overline{x}) = g(\overline{x})$, we obtain

$$
\max_{i=0,\dots,N-1}\mathbb{E}[|\delta_i|^2] \le \sum_{i=0}^{N-1} L^{N-1-i}M\varepsilon_i^{l,m,\theta^*} \le M\varepsilon^{l,m}\sum_{i=0}^{N-1}L^i = M\frac{L^N-1}{L-1}\varepsilon^{l,m}.
$$

□

B Appendix

Proof of (4.7). We first show the finiteness of the first term in (4.7) by induction, where $i=0$ obviously holds. Assume that it holds at level i, by Assumption 1, Hölder's inequality, the independence between B_h and $\overline{X}_{ih}^\pi$, and (4.1), we have

$$
\begin{aligned}
\mathbb{E}\Big[\max_{0\le k\le i+1}\|X_{i+1}^\pi(k)\|_d^q\Big] &\le \mathbb{E}\Big[\max\Big(\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q, \|X_{i+1}^\pi((i+1)h)\|_d^q\Big)\Big]\\
&= \mathbb{E}\Big[\max\Big(\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q,\ \|X_i^\pi(i)+b\big(ih,\overline{X}_{ih}^\pi\big)h+\sigma\big(ih,\overline{X}_{ih}^\pi\big)B_h\|_d^q\Big)\Big]\\
&\le \mathbb{E}\Big[\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q\Big] + \mathbb{E}\left[\|X_i^\pi(i)+b\big(ih,\overline{X}_{ih}^\pi\big)h+\sigma\big(ih,\overline{X}_{ih}^\pi\big)B_h\|_d^q\right]\\
&\le \mathbb{E}\Big[\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q\Big] + 3^q\mathbb{E}\Big[\|X_i^\pi(i)\|_d^q + h^q\big(K\left(|ih|^{1/2}+\|\overline{X}_{ih}^\pi\|\right)+\|b(0,(0)_{0\le s\le T})\|_d\big)^q\\
&\qquad +\|B_h\|_d^q\big(K\left(|ih|^{1/2}+\|\overline{X}_{ih}^\pi\|\right)+\|\sigma(0,(0)_{0\le s\le T})\|_{d\times d}\big)^q\Big]\\
&\le \mathbb{E}\Big[\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q\Big] + 9^q\mathbb{E}\left[\|X_i^\pi(i)\|_d^q\right] + h^q\left(K^q\left(T^{q/2}+\mathbb{E}\left[\|\overline{X}_{ih}^\pi\|^q\right]\right)+\|b(0,(0)_{0\le s\le T})\|_d^q\right)\\
&\qquad +\mathbb{E}\left[\|B_h\|_d^q\right]\left(K^q\left(T^{q/2}+\mathbb{E}\left[\|\overline{X}_{ih}^\pi\|^q\right]\right)+\|\sigma(0,(0)_{0\le s\le T})\|_{d\times d}^q\right)\\
&\le C_0 + C_1\mathbb{E}\Big[\max_{0\le k\le i}\|X_i^\pi(k)\|_d^q\Big] < \infty,
\end{aligned}
$$

where $C_0\ge 0$ and $C_1\ge 1$. In the second last inequality we used the fact that $\|\overline{X}_{ih}^\pi\|^q = \max_{0\le k\le i}\|X_i^\pi(k)\|_d^q$, and the centered Gaussian random variable B_h has finite $\mathbb{E}\left[|B_h|^q\right]$ for any choice of q. The finiteness of the second term in (4.7) is obvious since

$$
\mathbb{E}\left[\|\varphi(X_i^\pi)\|_k^q\right] \le K^q\mathbb{E}\big[\|\overline{X}_{ih}^\pi\|_{(i+1)d}^q\big] + \|\varphi(0,(0)_{0\le s\le T})\|_k^q < \infty.
$$

□

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